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Adaptive speed planning and cooperative formation operation system for special-purpose vehicles in field environments

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Abstract

In complex off-road environments, traditional operations of special-purpose vehicles often suffer from limited terrain perception, restricted speed regulation, and low formation coordination efficiency. This study proposes a systematic framework for autonomous field operations. We first develop a multi-source perception and control architecture that integrates GNSS, LiDAR, vision sensing, and onboard computing into a hierarchical ‘perception-decision-control’ closed loop, enabling both precise single-vehicle operation and coordinated multi-vehicle tasks. To mitigate roll and vibration arising from terrain-speed mismatch, we introduce an adaptive speed-planning method driven by terrain frequency-domain features. In addition, an improved distributed model predictive control for cooperative formation operation is designed, leveraging a leader-follower scheme and an articulated-vehicle model to achieve stable formation maneuvering and high-accuracy task execution. The system has been deployed on vibroseis vehicles and validated in real-world geological exploration, demonstrating strong robustness and engineering feasibility. Its successful operation at the exploration bases of the China National Petroleum Corporation further confirms its reliability and practical value.

1. Introduction

With the rapid advances in intelligent equipment, automatic control, and artificial intelligence, special-purpose vehicles (SPV) used in seismic exploration, disaster response, and other high-risk missions are increasingly expected to operate autonomously in complex off-road environments [1, 2]. However, most platforms still rely heavily on human drivers when navigating unstructured terrain in practice. Strong terrain-induced disturbances, limited onboard perception [3], and the need for precise task execution make manual operation labor-intensive and error-prone, restricting operational efficiency, degrading environmental adaptability, and increasing safety risks [4, 5].

The vibroseis vehicle is a representative SPV and a key platform for seismic exploration and geophysical data acquisition. During large-scale surveys, multiple vibroseis vehicles travel in convoys, position themselves at predefined source points (SPs), and execute controlled vibratory excitations, earning the nickname ‘the CT machine of the Earth’ [6]. In current practice, vibroseis vehicle operations are predominantly manual: drivers follow planned routes, align the vehicle at each SP by experience, and trigger excitation based on voice or radio commands. This mode of operation makes positioning accuracy, excitation synchrony, and formation coordination highly sensitive to human factors and terrain disturbances. As a result, it becomes difficult to suppress vehicle roll and vibration caused by terrain-speed mismatch, to maintain accurate cooperative alignment of multiple vehicles, and to guarantee efficient execution of dense task points. These limitations motivate the development of an autonomous vibroseis vehicle system that integrates terrain-adaptive speed planning with robust multi-vehicle cooperative formation operation (CFO) [7, 8].

Adaptive speed control plays a critical role in ensuring the safety, mobility, and operational performance of SPV [9]. To achieve reliable operation in complex environments, such vehicles must not only perceive terrain characteristics but also adjust their speed adaptively based on terrain information [10, 11]. Huang *et al* [12] proposed a multimodal terrain recognition framework that integrates vision and inertial measurement unit (IMU) data, demonstrating strong robustness using field datasets. Nampoothiri *et al* [13] employed machine-learning-based terrain classification combined with IMU-assisted navigation, while Chen *et al* [14] enhanced recognition accuracy and robustness by fusing onboard sensing with visual models. However, most existing methods still rely on piecewise speed adjustment strategies during velocity planning, which often lead to discontinuous or abrupt speed transitions, thereby limiting vehicle stability and ride comfort on uneven terrain. Establishing a continuous and nonlinear mapping between terrain roughness and vehicle speed remains essential for achieving smooth control and safe navigation.

To improve speed regulation performance in challenging environments, researchers have explored a variety of adaptive speed control strategies. Wang *et al* [15] utilized wheel-force sensing to capture wheel-terrain interaction characteristics, enabling coordinated speed control under complex terrain conditions. Gong *et al* [16] introduced a data-efficient iterative learning control approach for speed trajectory planning in heavy-vehicle platoons, demonstrating convergence and stability in simulation. Hu *et al* [17] developed an adaptive trajectory planning method that simultaneously accounts for friction, speed, safety, and comfort, achieving good feasibility under multiple constraints. Artuñedo *et al* [18] proposed a time-optimal speed planning approach subject to acceleration, velocity, and ride-quality constraints, validated in structured road environments. Sun *et al* [19] further employed model predictive control (MPC) to achieve adaptive speed tracking in intelligent connected vehicles. Despite these advances, most existing methods rely primarily on simulation or structured-road experiments and lack real-vehicle validation tailored to the dynamics and operational conditions of heavy-duty SPV in unstructured terrain.

Formation control of SPV is a key technology for enabling multi-vehicle cooperation and improving mission efficiency, with broad applications in mining transportation, emergency rescue, and unmanned swarm operations [20–22]. Most studies have focused on vehicle formation control in structured road scenarios. For instance, Peng *et al* [23] proposed a sliding-mode-based formation algorithm, achieving rapid convergence to a stable formation, but validation was limited to numerical simulations without real-vehicle experiments. Li *et al* [24] combined PID control with MPC to design a convoy-following controller and demonstrated its effectiveness in simulation. Wen *et al* [25] developed a CF-MPC controller inspired by human driving behavior and validated its effectiveness through theoretical analysis and simulation. Although these studies laid the foundation for multi-vehicle formation control, most approaches rely on simplified vehicle models and structured road assumptions, with insufficient consideration of complex terrains, heavy-load systems, and strong nonlinear couplings. Consequently, they exhibit notable limitations when applied to unstructured environments or heavy multi-vehicle systems.

To address these challenges, recent research has shifted toward MPC frameworks for heavy vehicles and unstructured environments. Xu *et al* [26] combined MPC with model-free adaptive control to develop a formation control framework for precise trajectory tracking of autonomous mining vehicles. Ozkan and Ma [27] designed a linear MPC controller accounting for dynamic losses within a vehicle formation, achieving an improved trade-off between speed smoothness and headway tracking and thereby enhancing fuel efficiency. Ard *et al* [28] further proposed a distributed stochastic MPC (DSMPC) scheme for heavy-duty vehicles, in which the lead vehicle learns human-driving behavior via stochastic inverse reinforcement learning and the follower vehicles employ DSMPC for autonomous platooning. Simulation results showed clear advantages over conventional distributed MPC. Despite these advances, vibroseis vehicles fundamentally differ from conventional heavy vehicle platoons in terms of large payloads, articulated chassis structures, and the requirements for coordinated multi vehicle operations. These characteristics introduce strong kinematic and dynamic couplings as well as pronounced nonlinearities, which are not explicitly addressed by existing formation control frameworks.

To address key challenges faced by SPV operating in unstructured environments, this study develops a systematic framework for autonomous SPV systems. The vibroseis vehicle is adopted as a representative SPV in this study and serves as the experimental platform for system design and validation. The main contributions are summarized as follows.

- (1) **Development of an integrated system architecture for autonomous vibroseis vehicles.** A multi-sensor perception and control framework is developed by integrating GNSS, LiDAR, vision sensors, and an onboard computing platform. The system has been deployed at the seismic

exploration base of the China National Petroleum Corporation, demonstrating strong engineering feasibility and operational effectiveness.

- (2) **An adaptive speed planning method based on frequency-domain terrain features.** A terrain roughness index is derived from the frequency characteristics of terrain point clouds, enabling the construction of a terrain-speed mapping model. This supports dynamic speed adjustment across varying terrain, improving stability and safety during operations in unstructured environments.
- (3) **A distributed MPC framework for CFOs of vibroseis vehicles.** Task points are allocated via convex quadratic programming, and local paths are refined with quintic Bézier curves, explicitly considering vehicle kinematics and operational constraints to achieve precise, feasible, and coordinated multi-vehicle operations in field environments.

The paper is structured as follows: section 2 introduces the autonomous vibroseis vehicle system architecture. Section 3 presents the terrain-roughness-based adaptive speed-planning method. Section 4 describes the improved distributed MPC framework for CFOs. Section 5 reports the experimental results, and section 6 concludes the paper.

2. System overview of the autonomous vibroseis vehicle

2.1. Vibroseis vehicle platform and mechanical architecture

The vibroseis vehicle is a specialized articulated engineering vehicle distinguished by its unique body structure, steering mechanism, and operational mode compared with conventional engineering vehicles. As illustrated in figure 1, the system mainly consists of four subsystems: the driving system, articulation system, operating system, and power system. The vehicle adopts a full hydraulic drive architecture, in which the power system supplies hydraulic energy to both the driving motors and steering cylinders, thereby realizing vehicle propulsion and steering control. The articulation system divides the vehicle into front and rear bodies connected by an articulated joint. Steering is achieved through the coordinated extension and retraction of two hydraulic cylinders located at the articulation point, enabling a flexible bending motion of the vehicle. The key geometric dimensions and main performance parameters of the vehicle are summarized in table 1.

2.2. Characteristics of vibroseis vehicles CFOs

The formation operation of vibroseis vehicles represents one of the most collaborative and technically demanding processes in seismic exploration. Its primary objective is to enable multiple vibroseis vehicles to move in a coordinated formation along a predefined trajectory and perform high-precision excitation tasks upon arrival at the designated work area, using the system's center of gravity as a reference, as illustrated in figure 2. In rugged off-road environments, surface undulations and terrain variations affect the stability and coordination of formation traveling. To ensure reliable motion, each vehicle continuously perceives ground conditions and assesses terrain roughness. Based on the terrain features ahead, the system adaptively adjusts the vehicle speed to maintain stability and preserve the formation geometry.

Once the formation enters the operational zone, each vehicle rapidly moves to its assigned work point, ensuring that the formation's SP is precisely aligned with the SP to satisfy operational accuracy requirements. This guarantees a uniform distribution of excitation energy within the target strata. After completing the excitation operation, the vehicles reassemble into the original formation and proceed to the next work area. In this process, the optimal inter-point spacing distance is 12 m, while the safe formation distance is 13 m, establishing a closed-loop cooperative workflow that integrates formation traveling, operational deployment, and formation recovery.

2.3. Hardware architecture

The overall hardware configuration of the autonomous driving system for the vibroseis vehicle is illustrated in figure 3. Functionally, the system comprises four major subsystems: the perception and localization unit, the control unit, the actuation unit, and the human-machine interface (HMI) unit.

The perception and localization unit integrates multiple onboard sensors, including LiDAR, millimeter-wave radar, cameras, steering angle sensors, and a GNSS navigation system. This unit provides the control system with real-time information on the surrounding environment, vehicle status, and precise pose estimation. The control unit consists of an onboard industrial computer and a platform for unified simulation and testing (PLUS1) controller. The onboard computer executes high-level algorithms such as perception, decision-making, and motion planning, while the PLUS1 controller governs the vehicle's powertrain and hydraulic subsystems to ensure real-time control performance. The

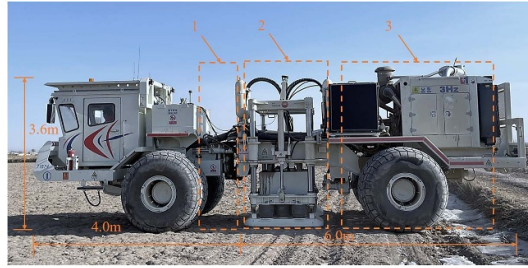


Figure 1. Body architecture of the vibroseis vehicle. 1. Articulated system. 2. Operating system. 3. Power system.

Table 1. Dimensions and performance parameters of the vibroseis vehicle.

Parameter	Values
Front vehicle body length	4.0 m
Rear vehicle body length	6.0 m
Vehicle width	3.38 m
Vehicle height	3.6 m
Total mass	32 000 kg
Design speed	0–26 km h ⁻¹
Articulation angle	–35°~35°
Maximum gradeability	40%

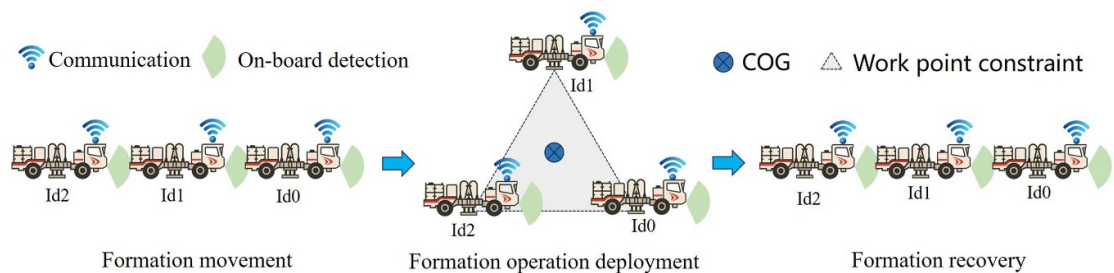


Figure 2. Schematic diagram of the vibroseis vehicle formation operation process.

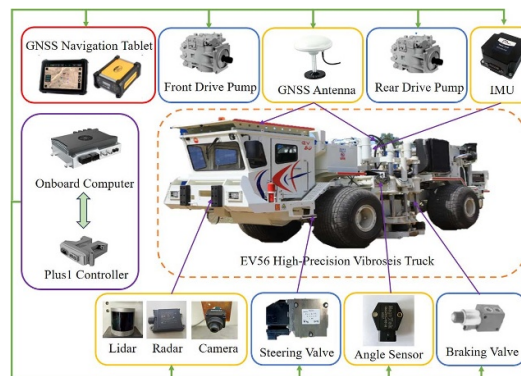


Figure 3. Illustration of the hardware architecture for the autonomous vibroseis vehicle system.

actuation the unit includes front drive pump, the rear drive pump, the steering valve, and the braking valve. The front and rear drive pumps provide the hydraulic power required for propulsion and torque distribution across varying terrain conditions, enabling stable and controllable vehicle motion. The steering valve directly receives pulse width modulation (PWM) signals from the controller to regulate hydraulic flow, thereby achieving accurate articulation angle control. The braking valve is responsible for emergency braking and hill-hold functions, ensuring system safety and reliability. The HMI unit consists

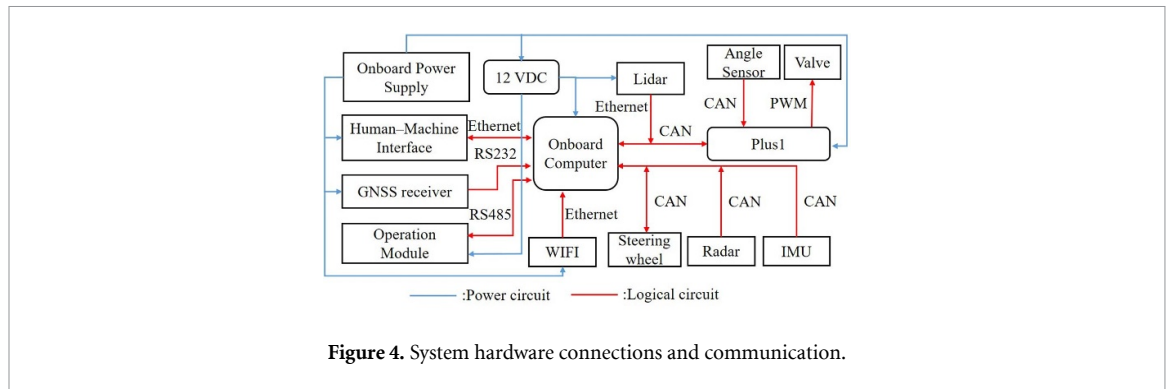


Figure 4. System hardware connections and communication.

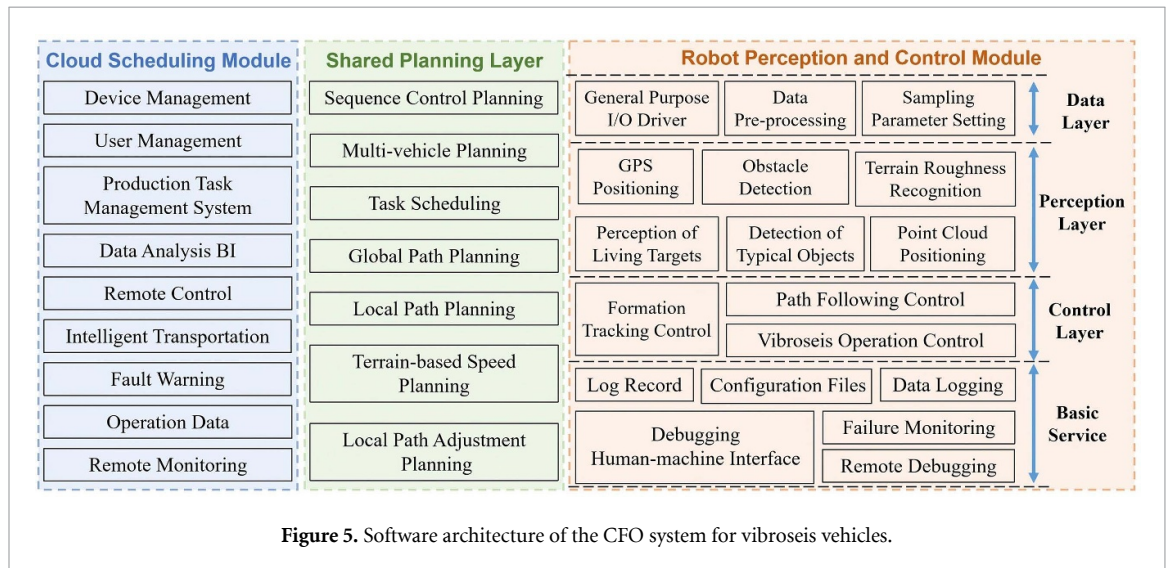


Figure 5. Software architecture of the CFO system for vibroseis vehicles.

of a GNSS-based navigation tablet that visualizes real-time operational information, including sensor health status, vehicle pose and environmental perception outputs.

As shown in figure 4, data communication among hardware modules is implemented via multiple interfaces, including Ethernet, controller area network (CAN) bus, RS-232, and RS-485. The onboard computer serves as the central node. Ethernet communication supports data exchange between the onboard computer and the HMI module. The CAN bus handles real-time communication with the PLUS1 controller, millimeter-wave radar, IMU, and electric steering system. RS-232 is used to receive raw GNSS data, while RS-485 enables communication with the excitation operation module. At the lowest level, the controller drives hydraulic actuators such as steering and braking valves through PWM signals to achieve precise vehicle control.

2.4. Software architecture

The CFOs system for vibroseis vehicles is developed primarily in C/C++, built on the Linux and ROS platforms, and deployed under Ubuntu 20.04. The system is implemented on ROS 1, where communication between ROS nodes is handled via topic-based message passing, with the control node publishing at 10 Hz. Inter-vehicle communication within the vibroseis fleet is facilitated through a self-organizing mesh radio network. As shown in figure 5, the software system adopts a multi-layer architecture consisting of a cloud scheduling module, a shared planning layer, and a vehicle perception and control module. Within the perception and control module, the architecture is further divided into the data layer, perception layer, control layer, and basic service layer, enabling full-chain functional support from low-level data acquisition to high-level decision execution. This hierarchical design enhances system scalability and provides a reliable software foundation for multi-vehicle cooperation in complex terrain environments.

The cloud scheduling module is responsible for remote management and multi-vehicle coordination. Its core functions include equipment and user management, production task allocation, operational data analysis and management, intelligent scheduling, remote monitoring, and fault diagnosis. Through unified management of vehicles and personnel, optimization of task scheduling strategies, and real-time

monitoring of vehicle states, this module ensures system safety and reliability during intensive field operations, serving as the upper-level support for autonomous fleet coordination.

The shared planning layer, together with the vehicle perception and control module, enables cooperative decision-making and real-time path tracking. The shared planning layer covers sequence control, multi-vehicle cooperative planning, global and local path generation, and terrain-adaptive velocity adjustment and local path adjustment for formation optimization. The perception and control module acquires and preprocesses sensory data through the data layer, performs vehicle localization, terrain roughness recognition, and obstacle detection at the perception layer, executes formation tracking and vibration operation control at the control layer, and provides logging, configuration, and remote debugging services through the basic service layer. This modular layered architecture ensures real-time performance and robustness, supporting high-precision and safe autonomous operations of multiple vehicles in complex terrain environments.

3. Adaptive speed planning based on frequency-domain terrain features

To enhance the stability and safety of vibroseis vehicle formations operating in complex environments, an adaptive speed planning algorithm based on frequency-domain terrain features is developed for the leading vehicle, as illustrated in figure 6. This method quantitatively evaluates terrain roughness by analyzing surface undulations using LiDAR point clouds and dynamically adjusts the leading vehicle's speed accordingly, thereby effectively preventing vehicle roll and vibration caused by mismatches between terrain and speed during formation movement.

3.1. Terrain roughness assessment

The forward-mounted LiDAR system of the vibroseis vehicle collects ground point cloud data within a look-ahead distance K . After filtering and ground segmentation, the longitudinal profile height sequence z along the vehicle's central axis is extracted. The sequence is then subjected to a fast Fourier transform (FFT) to analyze the terrain's frequency-domain characteristics:

$$y_c = \text{fft}(z). \quad (1)$$

The complex sequence y_c represents the amplitude characteristics of the input signal at each frequency, rather than the actual amplitude. To obtain the FFT amplitude $A(i)$ corresponding to the i th complex point $y_c(i)$, the following expression is used:

$$A(i) = \begin{cases} \frac{|y_c(i)|}{N}, & i = 0, \\ \frac{|y_c(i)|}{N/2}, & i = 1, 2, \dots, N-1, \end{cases} \quad (2)$$

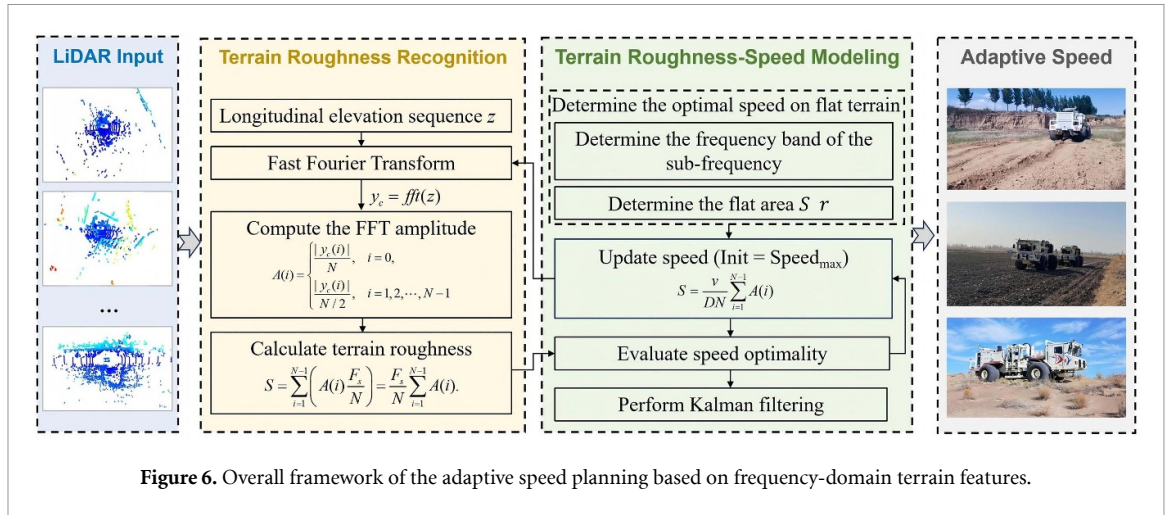
where N is the number of sampling points in the sequence z , and $|y_c(i)|$ denotes the modulus of the i th point. Since FFT does not directly provide the frequency distribution, the frequency resolution $\Delta f = F_s/N$ is determined using the sampling frequency F_s and the number of points N . The corresponding frequency range is given by:

$$F(i) = i \frac{F_s}{N}, \quad i = 0, 1, \dots, N-1. \quad (3)$$

A strong correlation exists between terrain roughness and the energy distribution of the longitudinal profile within the target sub-frequency band, which is commonly used to characterize ride-relevant road irregularities in frequency-domain roughness models [29]. Accordingly, the integral area under the sub-frequency band of the amplitude-frequency curve is adopted as a metric for terrain roughness. To mitigate interference from low-frequency DC components, the terrain roughness is defined as the area S enclosed by the amplitude-frequency curve (excluding the DC component) and the frequency axis, computed as:

$$S = \sum_{i=1}^{N-1} \left(A(i) \frac{F_s}{N} \right) = \frac{F_s}{N} \sum_{i=1}^{N-1} A(i). \quad (4)$$

This approach effectively quantifies terrain roughness and enables terrain classification based on the distribution of S , providing a quantitative characterization and recognition method.



3.2. Adaptive speed planning based on frequency-domain terrain features

To mitigate vehicle roll and vibration caused by mismatches between terrain roughness and driving speed, an adaptive speed-planning algorithm is developed based on the frequency-domain roughness index introduced in section 3.1. The key idea is to use the FFT-based roughness measure as a constraint and to determine, for each local terrain segment, the highest admissible speed that keeps the terrain-induced excitation within an acceptable range. It should be noted that the following formulation is intended as a linear approximation within the normal operating envelope of vibroseis vehicles, rather than a universal terrain–speed law.

From equation (4), for a given terrain profile the roughness index S is proportional to the sampling frequency F_s when the amplitude spectrum $A(i)$ is fixed. The temporal sampling frequency of the terrain signal is related to the vehicle speed v by

$$F_s = \frac{v}{D}, \quad (5)$$

where D is the spatial sampling interval. Substituting equation (5) into equation (4) yields the roughness index as a function of speed:

$$S(v) = \frac{v}{DN} \sum_{i=1}^{N-1} A(i), \quad (6)$$

where N is the number of sampling points. Equation (6) indicates that, for a fixed terrain segment and within a moderate roughness range, the roughness index exhibits an approximately linear dependence on vehicle speed. This linear relationship is adopted as a practical approximation to enable efficient real-time speed planning.

To establish a reference for safe and comfortable operation, the vehicle is first driven on a relatively flat terrain segment at a nominally comfortable high speed. The corresponding FFT amplitude spectrum is computed, and the roughness index evaluated from equation (6) is recorded as the reference roughness S^* . This value serves as a baseline threshold for subsequent speed planning on more rugged terrain.

During autonomous operation, a sliding window is applied to the forward-looking terrain profile to extract the current longitudinal height sequence. For each window, the FFT is computed to obtain $A(i)$, and a discrete set of candidate speeds is defined in the interval $[v_{\min}, v_{\max}]$ with a predefined step size Δv . For each candidate speed v in this set, the corresponding roughness index $S(v)$ is evaluated using equation (6). The optimal speed for the current terrain segment is then selected as the maximum candidate speed that satisfies

$$S(v) \leq S^*. \quad (7)$$

In this way, the algorithm realizes a continuous mapping from the local terrain roughness to the desired vehicle speed, ensuring that the excitation induced by terrain undulations does not exceed the reference level that was established on flat ground. It is emphasized that for extremely rough or highly irregular terrain profiles, nonlinear effects may become dominant, and such conditions fall outside the intended operational scope of the proposed method. Finally, the sequence of desired speeds obtained

from the sliding-window search is filtered using a Kalman filter to suppress measurement noise and avoid abrupt changes in the speed command.

4. Improved distributed MPC for CFOs

In complex and highly dynamic field environments, manual operation is inherently inefficient and cannot guarantee precise spatial alignment of vibroseis vehicles. Meanwhile, conventional distributed MPC approaches neglect the articulated kinematic characteristics and task-specific operational constraints of vibroseis vehicles. To address these challenges, this study proposes an improved distributed MPC for CFO of vibroseis vehicles. As illustrated in figure 7, the proposed framework adopts a two-layer architecture, consisting of a high-level planning layer and a low-level distributed MPC control layer. The distributed MPC explicitly incorporates the articulated kinematic model of vibroseis vehicles and enforces steering angle and articulation constraints to ensure motion feasibility.

In non-SP operational areas, the vehicle fleet follows a predefined global reference route. Upon entering the SP operational area, task points are assigned according to the spatial distribution of vibration points. To satisfy both operational efficiency and positioning accuracy, a two-stage strategy is adopted: first, convex quadratic programming is used to optimize the assignment of task points; subsequently, local path refinement is performed using quintic Bézier curves, generating smooth and dynamically feasible approach trajectories. The proposed algorithm covers the full workflow, including formation movement, deployment, and recovery, and achieves stable multi-vehicle cooperation and high-precision vibration-point positioning.

4.1. Kinematic modeling of vibroseis vehicles

Vibroseis vehicles belong to heavy-duty SPV, typically featuring an articulated front and rear body structure, which results in maneuverability and stability characteristics significantly different from those of conventional rigid vehicles. To accurately describe the motion characteristics of vibroseis vehicles, an articulated-vehicle kinematic model is adopted, as illustrated in figure 8. In this model, the state variables are defined as the longitudinal and lateral positions of the midpoint of the front axle (x_f, y_f) , the heading angle of the front body θ_f , and the articulation angle γ . The control inputs are the longitudinal velocity of the front body along its heading v_f and the articulation angular velocity ω . The resulting non-linear kinematic equations are expressed as:

$$\begin{pmatrix} \dot{x}_f \\ \dot{y}_f \\ \dot{\theta}_f \\ \dot{\gamma} \end{pmatrix} = \begin{pmatrix} \cos \theta_f \\ \sin \theta_f \\ \frac{-\sin \gamma}{l_f \cos \gamma + l_r} \\ 0 \end{pmatrix} v_f + \begin{pmatrix} 0 \\ 0 \\ \frac{-l_r}{l_f \cos \gamma + l_r} \\ 1 \end{pmatrix} \omega, \quad (8)$$

where l_f and l_r denote the distances from the front and rear body centroids to the articulation point, respectively. This model accurately captures the pose variation of vibroseis vehicles and provides a dynamic constraint foundation for subsequent control algorithms.

4.2. Task point optimization and local path correction planning

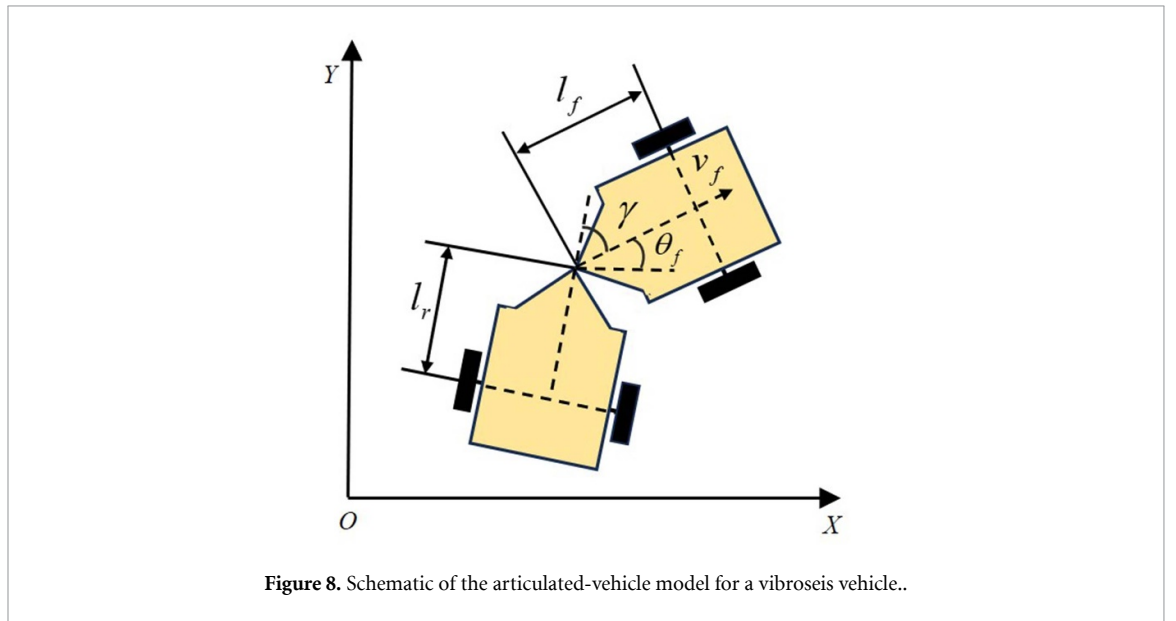
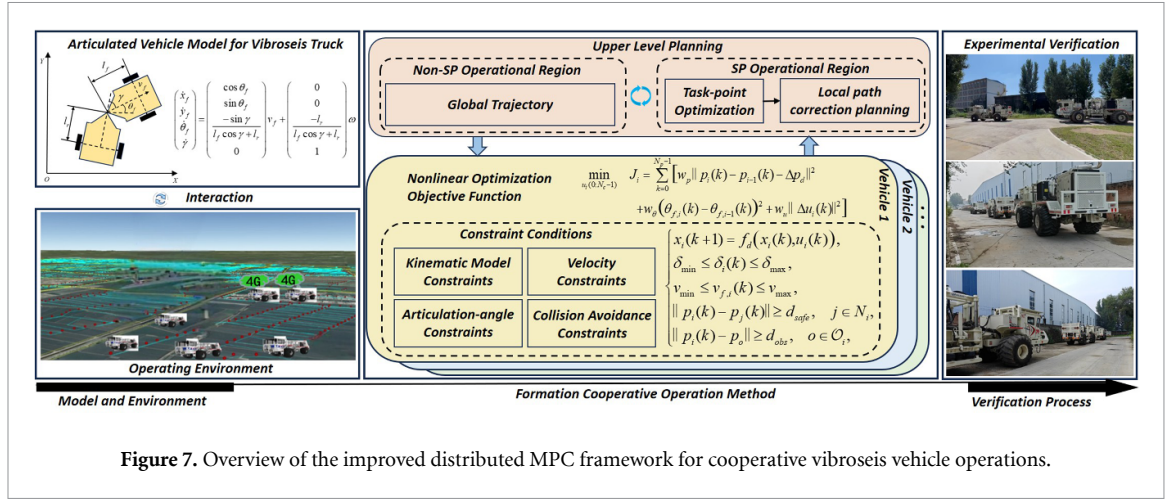
Upon entering the SP operation region, each vehicle determines the spatial layout of vibration points based on the location of the excitation center, ensuring that the geometric centroid of all assigned points is precisely aligned with the SP. By integrating the current vehicle states, the system achieves rapid deployment and accurate operational point positioning. To enhance operational efficiency and trajectory smoothness, a two-stage strategy is adopted: task point optimization using convex quadratic programming [30], followed by local path-correction planning based on quintic Bézier curves [31] to generate smooth and dynamically feasible approach trajectories.

Let the formation consist of N vehicles with current positions $p_i = [x_i, y_i]^T$, an operation center at $p_{SP} = [x_{SP}, y_{SP}]^T$, and assigned task points $q_j = [x_j, y_j]^T$. The task points are required to satisfy the geometric center constraint with respect to the SP:

$$\left\| \frac{1}{N} \sum_{j=1}^N q_j - p_{SP} \right\| \leq d. \quad (9)$$

Additionally, to avoid spatial interference among task points, a minimum safe distance constraint is imposed:

$$\|q_i - q_j\| \geq d_{\text{safe}}. \quad (10)$$



The task point optimization is formulated as the following optimization problem:

$$\begin{aligned}
 \min \quad & \sum_{i=1}^M \sum_{j=1}^N \|p_i - q_j\|^2 \\
 \text{s.t.} \quad & \left\| \frac{1}{N} \sum_{j=1}^N q_j - p_{\text{SP}} \right\| \leq d, \\
 & \|q_i - q_j\| \geq d_{\text{safe}}, \quad \forall i \neq j.
 \end{aligned} \tag{11}$$

This optimization problem employs convex quadratic programming [30] to determine the optimal task points for each vehicle. Subsequently, to enable a smooth transition of the vehicles from their current positions to the assigned task points, quintic Bézier curves are used to generate local correction paths. The curves guarantee C^3 continuity, allowing constraints on position, velocity, and acceleration at both the start and end points. Let the start and end points be $p_i = [x_i, y_i]^T$ and $q_j = [x_j, y_j]^T$, respectively. The path is expressed as:

$$B(t) = \sum_{k=0}^5 \binom{5}{k} (1-t)^{5-k} t^k z_k, \quad t \in [0, 1], \tag{12}$$

where z_k denotes the set of control points, determined by imposing boundary conditions on the vehicle's pose and curvature continuity.

After completing the tasks, the vehicles follow reversed quintic spline paths to return to the original formation, forming a closed-loop cooperative operation workflow.

4.3. Distributed MPC for path tracking

To maintain geometric consistency and ensure safe inter-vehicle spacing during trajectory tracking, a distributed MPC algorithm for CFO is developed. Unlike conventional DMPC approaches, it explicitly incorporates the articulated kinematic model of vibroseis vehicles and enforces steering angle and articulation-related feasibility constraints, enabling dynamically consistent motion planning. The controller adopts a leader-follower communication topology, incorporating the state and constraint information of neighboring vehicles into each local optimization problem. This formulation preserves the decentralized structure of individual controllers while enabling coherent, coordinated behavior across the entire formation.

Let the state vector of the i th vehicle be $x_i = [x_{f,i}, y_{f,i}, \theta_{f,i}, \gamma_i]^T$, and the control input be $u_i = [v_{f,i}, \omega_i]^T$. Using equation (8), the discrete-time kinematic model under sampling period T_s is expressed as:

$$x_i(k+1) = f_d(x_i(k), u_i(k)), \quad (13)$$

where $f_d(\cdot)$ denotes the discrete dynamic model of the vibroseis vehicle.

The nonlinear optimization objective for the i th vehicle is defined as

$$\min_{u_i(0:N_c-1)} J_i = \sum_{k=0}^{N_p-1} [w_p \|p_i(k) - p_{i-1}(k) - \Delta p_d\|^2 + w_\theta (\theta_{f,i}(k) - \theta_{f,i-1}(k))^2 + w_u \|\Delta u_i(k)\|^2], \quad (14)$$

where w_p , w_θ , and w_u denote the weighting coefficients. N_p and N_c denote the prediction and control horizons, respectively; $p_i = [x_{f,i}, y_{f,i}]^T$ is the vehicle position; and Δp_d represents the desired inter-vehicle displacement to preserve the formation geometry.

The optimization is subject to the following constraints:

$$\begin{cases} x_i(k+1) = f_d(x_i(k), u_i(k)), \\ \delta_{\min} \leq \delta_i(k) \leq \delta_{\max}, \\ v_{\min} \leq v_{f,i}(k) \leq v_{\max}, \\ \|p_i(k) - p_j(k)\| \geq d_{\text{safe}}, \quad j \in N_i, \\ \|p_i(k) - p_o\| \geq d_{\text{obs}}, \quad o \in \mathcal{O}_i, \end{cases} \quad (15)$$

where δ denotes the steering angle, N_i denotes the neighbor set of vehicle i , d_{safe} denotes the minimum inter-vehicle safety distance, and $\mathcal{O}_i$ is the set of obstacles within the prediction horizon of vehicle i , with p_o specifying the obstacle center position. The optimization constraints encompass vehicle dynamics, steering angle, velocity, inter-vehicle safety distance, and obstacle avoidance, ensuring safe and stable formation operation in field environments.

The proposed improved distributed MPC framework for CFO of vibroseis vehicles ensures formation stability, achieves precise path tracking, and satisfies the operational requirements of vibroseis vehicle formations.

5. Experiments and data analysis

5.1. Performance evaluation of the adaptive speed planning

To evaluate the performance of the proposed adaptive speed-planning method, two sets of experiments are designed. The first set investigates the performance of the proposed method in real experimental environments, while the second set provides a comparative evaluation with different speed-planning methods.

5.1.1. Experimental evaluation of adaptive speed planning in real environments

To validate the effectiveness of the proposed adaptive speed planning algorithm based on terrain frequency-domain features, a series of comparative experiments were conducted on a vibroseis vehicle platform. Two representative surface conditions were selected for testing: flat terrain and rugged terrain, as illustrated in figure 9. Each terrain segment was configured as a 30 m test track. During traversal, the vehicle continuously adjusted its desired speed based on the real-time estimation of the terrain roughness index S , with the allowable speed range constrained between 1.2 and 3.0 m s⁻¹. In the roughness

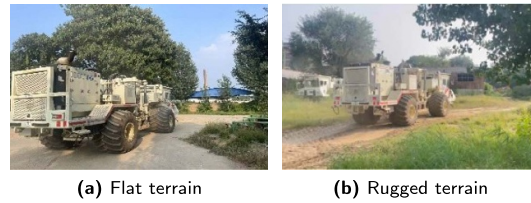
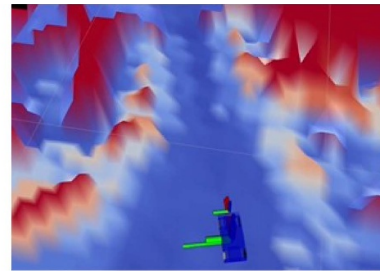
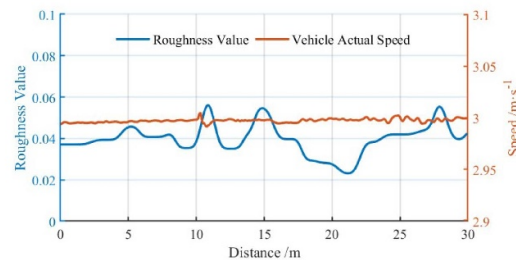


Figure 9. Typical terrain conditions selected for experimental evaluation.



(a) Estimated roughness map



(b) Vehicle speed adaptation profile

Figure 10. Roughness evaluation and corresponding speed adaptation profile for the flat-terrain test.

visualization maps, warmer colors (red) indicate stronger terrain undulations, whereas cooler colors (blue) correspond to flatter surfaces.

Figure 10 presents the experimental results of the speed adaptation planning for the vibroseis vehicle operating on flat terrain. As shown in figure 10(b), the roughness index remains within a low and stable range of 0.02–0.06 along the entire 30 m segment, indicating that the surface undulation is minimal. Under this condition, the vehicle maintains the steady driving speed of 3 m s^{-1} without noticeable fluctuations. These results confirm the effectiveness and stability of the proposed adaptive speed-planning algorithm when operating in flat-terrain environments.

Figure 11 presents the experimental results of the adaptive speed-planning algorithm for the vibroseis vehicle operating on rugged terrain. As shown in figure 11(b), the terrain roughness varies within the range of 0.09–0.21 along the test segment, exhibiting a clear trend of surface undulation. Beginning at approximately 3.6 m, the roughness index gradually increases, indicating an intensification of terrain irregularities. Correspondingly, the vehicle's actual speed decreases progressively as the roughness rises. When the vehicle reaches around 13.2 m, the roughness starts to decline, resulting in a gradual increase in speed. Subsequently, near 21.9 m, the roughness increases again, and the vehicle speed correspondingly decreases. Overall, the speed trajectory adapts dynamically to the varying roughness levels, further validating the effectiveness of the proposed strategy under rugged terrain conditions.

5.1.2. Experimental comparison of different speed-planning methods

To evaluate the performance of the proposed algorithm, comparative experiments were conducted in rugged terrain environments against representative baseline strategies. Two used speed-planning approaches were selected for comparison. (a) Constant-speed strategy (CS): the vehicle travels at a fixed predefined speed regardless of terrain conditions. This strategy serves as a baseline to assess the necessity of terrain-aware speed adaptation. (b) Time-domain terrain-adaptive speed planning (TC-SA) [32]:

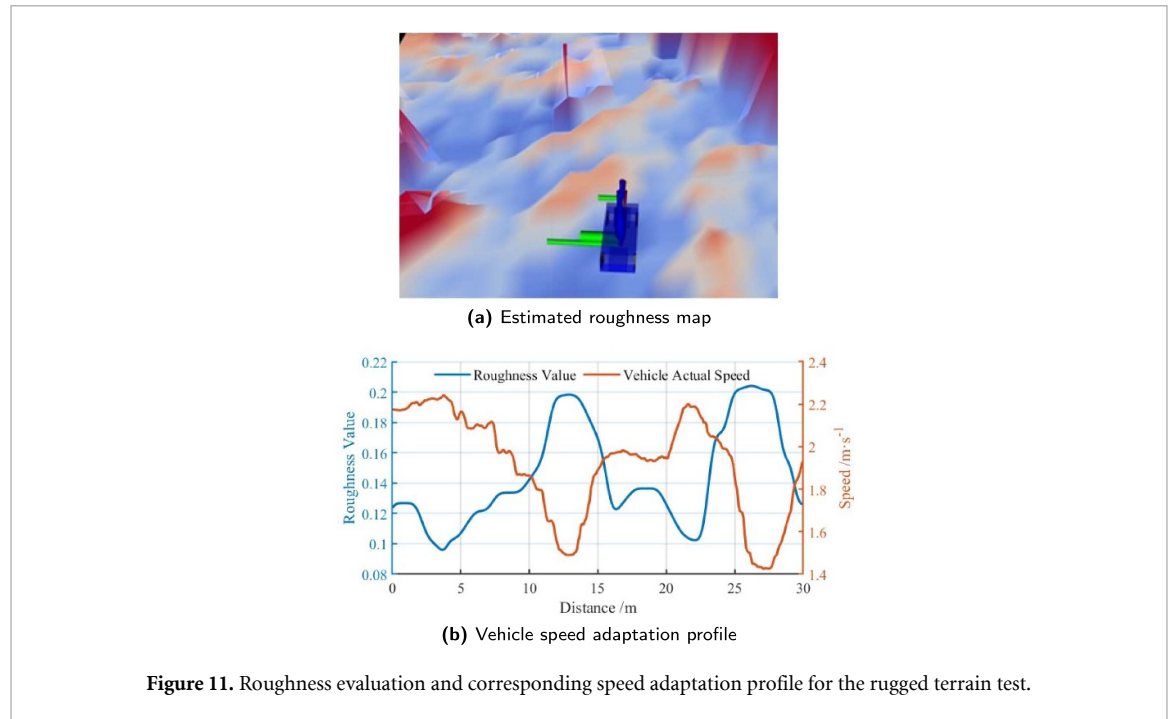


Figure 11. Roughness evaluation and corresponding speed adaptation profile for the rugged terrain test.

Table 2. Performance comparison of different speed-planning methods.

Method	RMS roll Angle (deg)	Max roll Angle (deg)	RMS vertical Acc. (m s^{-2})	Avg. Speed (m s^{-1})
CS	4.62 ± 0.38	6.00 ± 0.44	2.85 ± 0.31	2.98 ± 0.12
TC-SA	3.41 ± 0.29	5.20 ± 0.37	2.10 ± 0.26	2.25 ± 0.15
Ours	2.76 ± 0.21	5.50 ± 0.32	1.68 ± 0.19	2.53 ± 0.10

the vehicle adjusts its speed based on time-domain terrain features, representing a conventional terrain-adaptive approach without frequency-domain analysis. Besides, to evaluate vehicle stability, ride comfort, and operational efficiency, the following performance metrics are employed: (1) RMS roll angle, which quantifies lateral stability, where lower values indicate improved roll stability; (2) Maximum roll angle, reflecting extreme roll behavior and the associated safety margin; (3) RMS vertical acceleration, used to assess ride comfort and vibration suppression capability; (4) Average speed, representing overall operational efficiency. To mitigate the effects of randomness, each algorithm is executed five times, and the mean and standard deviation of the results are reported. The experimental results are summarized in table 2.

As shown in the table, the proposed method achieves the lowest RMS roll angle of 2.76° , corresponding to reductions of 1.86° and 0.65° compared with CS and TC-SA, respectively, which indicates a significant improvement in lateral vehicle stability. Although its maximum roll angle is slightly higher than that of TC-SA, it remains lower than the CS baseline and within safe operating limits. In terms of ride comfort, the proposed method attains the lowest RMS vertical acceleration of 1.68 m s^{-2} , demonstrating excellent vibration suppression capability. Meanwhile, it maintains a competitive average speed of 2.53 m s^{-1} , which is higher than that of TC-SA and only marginally lower than the aggressive CS strategy. Overall, the results confirm that the proposed adaptive speed planning method improves vehicle stability while maintaining high operational efficiency in rugged terrain environments.

In summary, the proposed adaptive speed-planning algorithm based on terrain frequency-domain features can effectively adjust vehicle speed in real time according to terrain complexity. This capability provides a robust and practical solution for intelligent speed regulation and terrain-adaptive control in vibroseis vehicle formations as well as other off-road vehicles.

5.2. Performance evaluation of CFO

To evaluate the performance of the proposed CFO method, two sets of experiments were designed. The first set investigates the performance of the proposed method under real experimental conditions. As no prior studies in the literature have addressed cooperative operations based on the characteristics of

Table 3. Performance of the CFO algorithm under different SP operation segments and formation sizes (mean $\pm$ standard deviation).

SP operation segment	Vehicles	Mean lateral Error (m)	Mean heading Error (rad)	Avg. Max lateral Error (m)	SP positioning Error (m)
Straight-line	2	0.143 ± 0.011	0.021 ± 0.003	0.382 ± 0.026	0.176 ± 0.015
	3	0.158 ± 0.013	0.027 ± 0.003	0.407 ± 0.030	0.254 ± 0.018
	4	0.172 ± 0.016	0.033 ± 0.005	0.411 ± 0.033	0.310 ± 0.023
Curved	2	0.157 ± 0.014	0.029 ± 0.004	0.385 ± 0.029	0.215 ± 0.019
	3	0.183 ± 0.018	0.036 ± 0.005	0.431 ± 0.036	0.286 ± 0.022
	4	0.219 ± 0.021	0.043 ± 0.007	0.463 ± 0.042	0.357 ± 0.028

seismic source vehicle formations, the second set of experiments focuses on evaluating the impact of different path-tracking algorithms on the performance of CFO.

5.2.1. Experimental evaluation of CFO in real environments

To assess the real-world performance of the improved distributed MPC framework for CFO of vibroseis vehicles, we conducted experiments under two representative SP-operation segments: straight-line segments and curved segments. The straight-line section evaluates the system's steady-state tracking capability under low-dynamic conditions, whereas the curved section examines its ability to maintain path alignment and task accuracy under varying curvature and strong inter-vehicle coupling. Formation sizes were set to 2–4 vehicles, and each configuration was tested in 10 repeated trials to obtain statistically meaningful results.

To quantitatively evaluate the performance of the CFO algorithm, four metrics were considered. The mean lateral tracking error and mean heading error were used to assess path-alignment accuracy and pose consistency, respectively. The average maximum lateral error captures the upper bound of trajectory deviation under operational conditions, while the SP positioning error is defined as the geometric deviation between the actual excitation point and the SP. The performance of the proposed algorithm was evaluated using the mean and standard deviation of each metric, reflecting the accuracy and stability of the CFO algorithm across repeated trials. The DMPC optimization problem was solved using the interior point optimizer [33] with a prediction horizon of 25 steps and a control horizon of 10 steps. Vehicle states and control inputs were shared via the onboard mesh communication system in real time. Table 3 summarizes the experimental performance across different road geometries and formation sizes.

From table 3, it can be observed that the improved distributed MPC framework for CFO consistently demonstrates strong tracking performance across different operation segments and formation sizes, while ensuring high SP operational accuracy. In the straight-line SP operation segment, the average lateral error remains below 0.18 m and the average heading error stays under 0.035 rad, showing only a slight increase as the number of vehicles grows. These results indicate that the framework can reliably maintain formation geometry and reproduce the desired trajectory in straight-line SP operation scenarios. Meanwhile, the SP positioning accuracy is within 0.17–0.31 m, confirming that the task point optimization and local correction path strategy can robustly converge the vehicles to their assigned SP points and meet practical operational accuracy requirements.

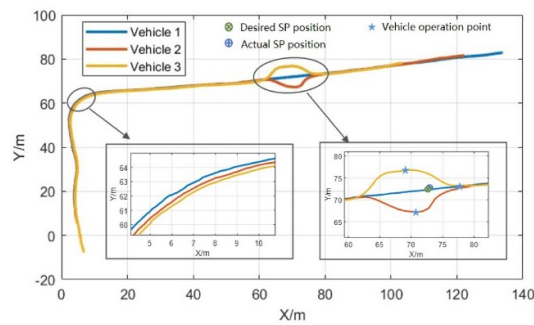
In curved SP operation segments, the increased curvature introduces additional coordination difficulty, leading to slightly higher errors: the average lateral error reaches up to 0.219 m, and the average heading error rises to 0.043 rad. Nevertheless, the overall performance remains stable and well-controlled, with the maximum lateral deviation not exceeding 0.47 m and no instability observed. The SP positioning error increases moderately with more vehicles, reaching 0.357 m, which still reflects satisfactory formation operation accuracy. Overall, the improved DMPC framework achieves stable path tracking and reliable SP positioning accuracy in both straight and curved terrain segments.

Figures 12 and 13 present the cooperative operation experiment using three vibroseis vehicles at a straight-line SP operation segment. Figure 12 illustrates the vehicles' real-world operating states at different stages, whereas figure 13 shows the top-down perspective and stereoscopic perspective trajectories as well as the heading evolution of each vehicle, where the X and Y axes represent the global horizontal and vertical coordinates of the multi-vehicle operational area.

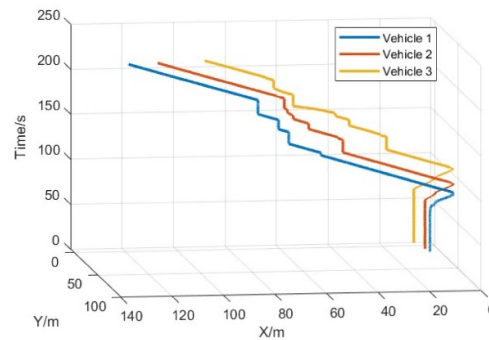
As shown in figure 13, during the formation-movement phase, the three-vehicle formation initially proceeds in a line-abreast configuration. After $t = 62.7$ s, the vehicles enter the SP operation zone and begin the coordinated deployment phase. Each vehicle moves toward its assigned SP point following a locally corrected path. Upon reaching the designated positions, the vehicles perform the required seismic tasks. The zoomed-in view in figure 13(a) shows that the actual SP position (72.7, 72.8) closely matches



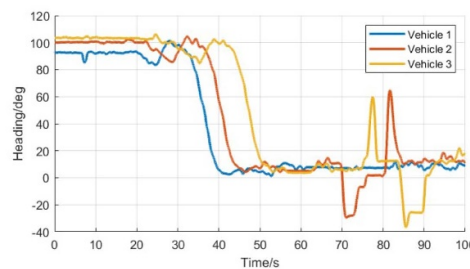
Figure 12. Three-vehicle formation operation process in the real-world environment. (a) Formation-movement phase. (b) Formation-operation deployment phase.



(a) Actual vehicle trajectories from the top-down view



(b) Actual vehicle trajectories from the stereoscopic view



(c) Heading evolution of each vehicle within the formations

Figure 13. Experimental results for a three-vehicle formation at a straight-line SP operation segment.

the desired SP position (72.5, 72.5), confirming the effectiveness of the SP layout and the local correction path planning. After completing the task at $t = 80.3$ s, the vehicles follow the corrected paths, reform the formation by $t = 92.8$ s, and then proceed to the next SP point.

Furthermore, figures 13(a) and (b) show that during formation motion, all three vehicles consistently track the reference trajectory: the leader follows the global path, and the followers accurately reproduce its motion, maintaining coherent formation geometry. The heading curves in figure 13(c) indicate highly stable orientation in straight segments and rapid yet well-damped adjustments in curved segments, demonstrating the robustness of the proposed framework in handling attitude transitions.

Figure 14 illustrates the distribution of positioning errors for 60 actual SP operation points during formation cooperative tasks. The horizontal axis represents the sequence of tested points, and the vertical axis denotes the deviation between the actual and desired SP positions. The results indicate an average

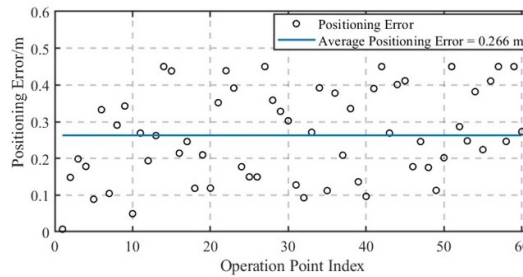


Figure 14. SP positioning errors during formation cooperative tasks..

Table 4. Performance comparison with different algorithms.

Method	Avg. Lateral Error (m)	Min. Inter-vehicle Distance (m)	Avg. SP positioning Error (m)
PID	0.42 ± 0.06	12.36 ± 0.45	0.48 ± 0.07
LQR	0.31 ± 0.04	12.84 ± 0.37	0.36 ± 0.04
Ours	0.18 ± 0.03	13.14 ± 0.22	0.27 ± 0.02

SP positioning error of 0.266 m, with all deviations constrained within 0.45 m, fully satisfying the design requirement of a maximum SP positioning error of 0.5 m.

5.2.2. Experimental comparison of path-tracking algorithms

To further evaluate the path-tracking capability of the proposed framework, comparative experiments were conducted against PID [34] and LQR [35] algorithms, which are widely adopted in practical vehicle systems. For a fair comparison, all tracking algorithms follow the same reference trajectories generated by the high-level planner, including global path segments and local quintic Bézier trajectories within the SP operation area. For PID and LQR, explicit safety distance constraints are enforced to guarantee collision-free operation. Both baseline algorithms are implemented based on a kinematic bicycle model rather than the articulated vehicle model.

To quantitatively evaluate performance, the following metrics are considered: (1) Average lateral tracking error, measuring overall path-following accuracy; (2) Minimum inter-vehicle distance, reflecting the level of safety during formation following; (3) Average SP positioning error, indicating task execution precision at vibration points. To mitigate the influence of randomness, each algorithm is executed five times, and the mean and standard deviation of the results are reported. The experimental results are summarized in table 4.

As shown in table 4, the proposed method achieves the smallest average lateral error of 0.18 m, representing reductions of 0.24 m and 0.13 m compared with PID and LQR, respectively. This improvement highlights the benefit of explicitly modeling articulated vehicle kinematic coupling and incorporating predictive optimization. In terms of formation safety, the proposed approach maintains the largest minimum inter vehicle distance of 13.14 m, indicating improved formation safety during formation following process. Furthermore, the proposed algorithm yields the lowest average SP positioning error of 0.27 m, outperforming PID and LQR by 0.21 m and 0.09 m, respectively, which is critical for precise vibroseis operations. Overall, the proposed distributed MPC framework consistently demonstrates superior tracking accuracy, enhanced formation safety, and improved task execution precision in complex field environments.

In summary, the proposed improved distributed MPC framework for CFO demonstrates excellent cooperative tracking performance and operational positioning accuracy. It ensures high SP positioning accuracy while exhibiting superior performance in trajectory tracking stability, formation deployment consistency, and attitude control. These experimental results provide strong support for high-precision autonomous operations of vibroseis vehicle formations.

6. Conclusion

This study addresses the limitations of traditional SPV operations in off-road environments, including insufficient manual operation accuracy, limited speed adaptation capability, and low formation coordination efficiency. We propose a systematic solution for field operations of SPV. The framework integrates

onboard multi-source perception and control, a cloud-based scheduling system, and a robotic operating system, effectively replacing traditional manual workflows in geological exploration. In addition, a terrain-frequency-based adaptive speed planning method is introduced. By performing spectral analysis on ground point clouds, a roughness index is extracted and mapped to vehicle speed, enabling smooth and dynamic speed adjustment across different terrain conditions. Building on this, an improved distributed MPC framework for CFOs was further developed. By integrating a leader–follower structure with an articulated-vehicle kinematic model, the algorithm enables more stable formation control and higher task accuracy in multi-vehicle operations.

Furthermore, the system has been deployed on vibroseis vehicles for geological exploration within the China National Petroleum Corporation, and undergone extensive field tests across multiple exploration bases. In a five-day comparative trial involving eight vehicles divided into two groups—one using autonomous formation control and the other manually driven—the autonomous group completed 1032 task points with an average positional error of 0.43 m, whereas the manual group completed 816 points with an average error of 1.36 m. Overall, the autonomous system improved average operational efficiency by approximately 20.9% and task-point accuracy by 68.3%, demonstrating significant practical advantages and engineering potential. In practical operation, the end-to-end control-command latency, from MPC output generation to vehicle actuation, was measured at an average of 16.4 ms. During multi-vehicle cooperative operations, the packet loss rate remained below 0.8%. These quantitative results indicate that, under practical operating conditions, the proposed ROS-based architecture is capable of reliably supporting real-time cooperative formation control tasks.

Although the proposed system has demonstrated significant improvements in operational efficiency and task accuracy, there remains room for further optimization. In particular, system performance may degrade under exceptional conditions such as severe terrain variations, extreme weather disturbances, or partial sensor failures, which could adversely affect formation coordination accuracy and control reliability. Future work will focus on incorporating terrain-informed constraints into the cooperative control framework, including static feasibility constraints related to terrain accessibility and dynamic stability constraints related to sideslip and rollover limitations, to enhance operational safety and precision in challenging environments. In addition, multi-sensor fusion techniques combined with deep reinforcement learning will be investigated to improve system robustness and intelligence under uncertain environmental conditions and sensing degradation.

Data availability statement

The data cannot be made publicly available upon publication because they contain sensitive personal information. The data that support the findings of this study are available upon reasonable request from the authors.

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